

Name: Sujay Sunil Mane
Roll No.: 31010924149
Class: TY BSc IT (B)
Subject: Web Services

Practical - 6

Tomcat



Server Status

Manager							
List Applications	HTML Manager Help	Manager Help	Complete Server Status				
Server Information							
Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/10.1.57	17.0.12+8-LTS-286	Oracle Corporation	Windows 11	10.0	amd64	ASD-LAB2-21	172.21.3.288

JVM

Free Memory: 84.06 MB Total Memory: 128.00 MB Max Memory: 256.00 MB

Memory Pool	Type	Initial	Total	Maximum	Used
G1 Eden Space	Heap memory	25.00 MiB	75.00 MiB	-0.00 MiB	30.00 MiB
G1 Old Gen	Heap memory	103.00 MiB	48.00 MiB	256.00 MiB	6.27 MiB (3%)
G1 Survivor Space	Heap memory	0.00 MiB	5.00 MiB	-0.00 MiB	4.20 MiB
CodeHeap 'non-nmethods'	Non-heap memory	2.43 MiB	3.37 MiB	7.31 MiB	1.77 MiB (24%)
CodeHeap 'non-profiled nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	116.37 MiB	1.86 MiB (1%)
CodeHeap 'profiled nmethods'	Non-heap memory	2.43 MiB	9.06 MiB	116.31 MiB	8.85 MiB (7%)
Compressed Class Space	Non-heap memory	0.00 MiB	2.56 MiB	1024.00 MiB	2.39 MiB (0%)
Metaspace	Non-heap memory	0.00 MiB	25.75 MiB	-0.00 MiB	25.35 MiB

"http-nio-8080"

Manager							
List Applications	HTML Manager Help	Manager Help	Complete Server Status				
Server Information							
Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/10.1.57	17.0.12+8-LTS-286	Oracle Corporation	Windows 11	10.0	amd64	ASD-LAB2-21	172.21.3.288

JVM

Free Memory: 84.06 MB Total Memory: 128.00 MB Max Memory: 256.00 MB

Memory Pool	Type	Initial	Total	Maximum	Used
G1 Eden Space	Heap memory	25.00 MiB	75.00 MiB	-0.00 MiB	30.00 MiB
G1 Old Gen	Heap memory	103.00 MiB	48.00 MiB	256.00 MiB	6.27 MiB (3%)
G1 Survivor Space	Heap memory	0.00 MiB	5.00 MiB	-0.00 MiB	4.20 MiB
CodeHeap 'non-nmethods'	Non-heap memory	2.43 MiB	3.37 MiB	7.31 MiB	1.77 MiB (24%)
CodeHeap 'non-profiled nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	116.37 MiB	1.86 MiB (1%)
CodeHeap 'profiled nmethods'	Non-heap memory	2.43 MiB	9.06 MiB	116.31 MiB	8.85 MiB (7%)
Compressed Class Space	Non-heap memory	0.00 MiB	2.56 MiB	1024.00 MiB	2.39 MiB (0%)
Metaspace	Non-heap memory	0.00 MiB	25.75 MiB	-0.00 MiB	25.35 MiB

"http-nio-8080"

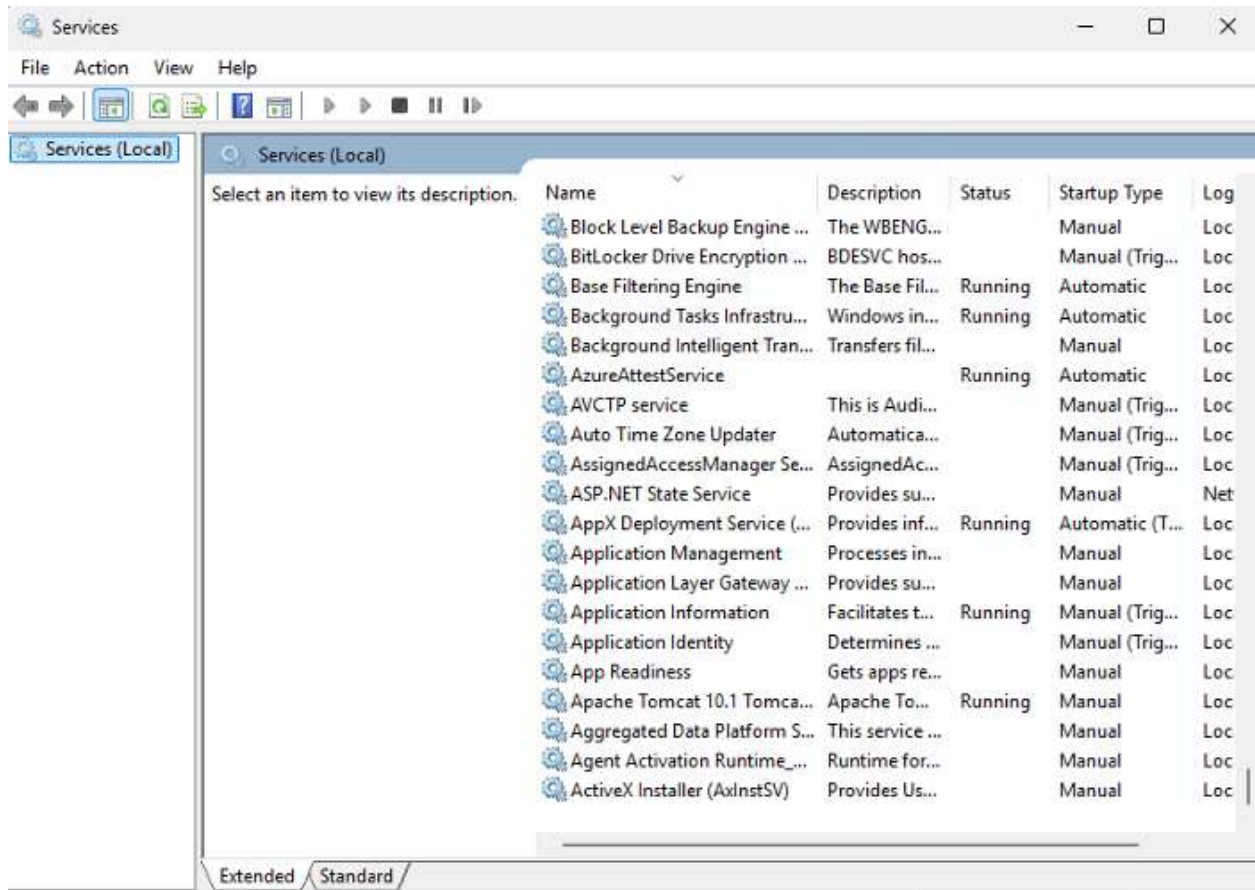
Max threads: 200 Current thread count: 10 Current threads (user): 1 Keep alive sockets count: 2
Max processing time: 648 ms Processing time: 1.029 s Request count: 10 Error count: 1 Bytes received: 0.00 MB Bytes sent: 0.05 MB

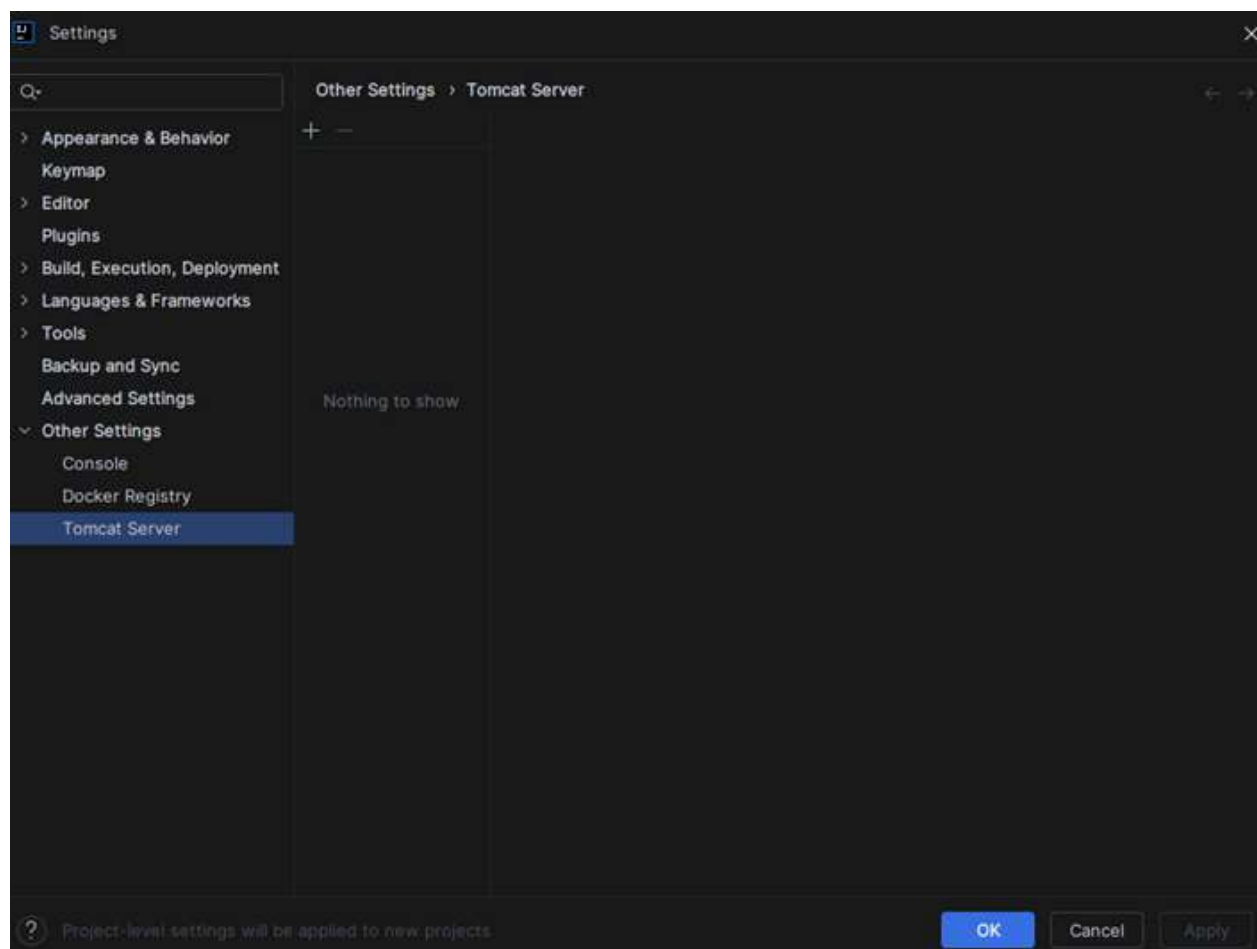
Stage	Time	Bytes Sent	Bytes Recv	Client (Forwarded)	Client (Actual)	VHost	Request
R	?	?	?	?	?	?	
R	?	?	?	?	?	?	
R	?	?	?	?	?	?	
S	37 ms	0 KB	0 KB	0 0 0 0 0 0 1	0 0 0 0 0 0 1	localhost	GET /manager/status HTTP/1.1
R	?	?	?	?	?	?	

P: Parse and prepare request S: Service F: Finish R: Ready K: Keepalive

Copyright © 1999-2020, Apache Software Foundation

Services





Source Code:

Index.jsp

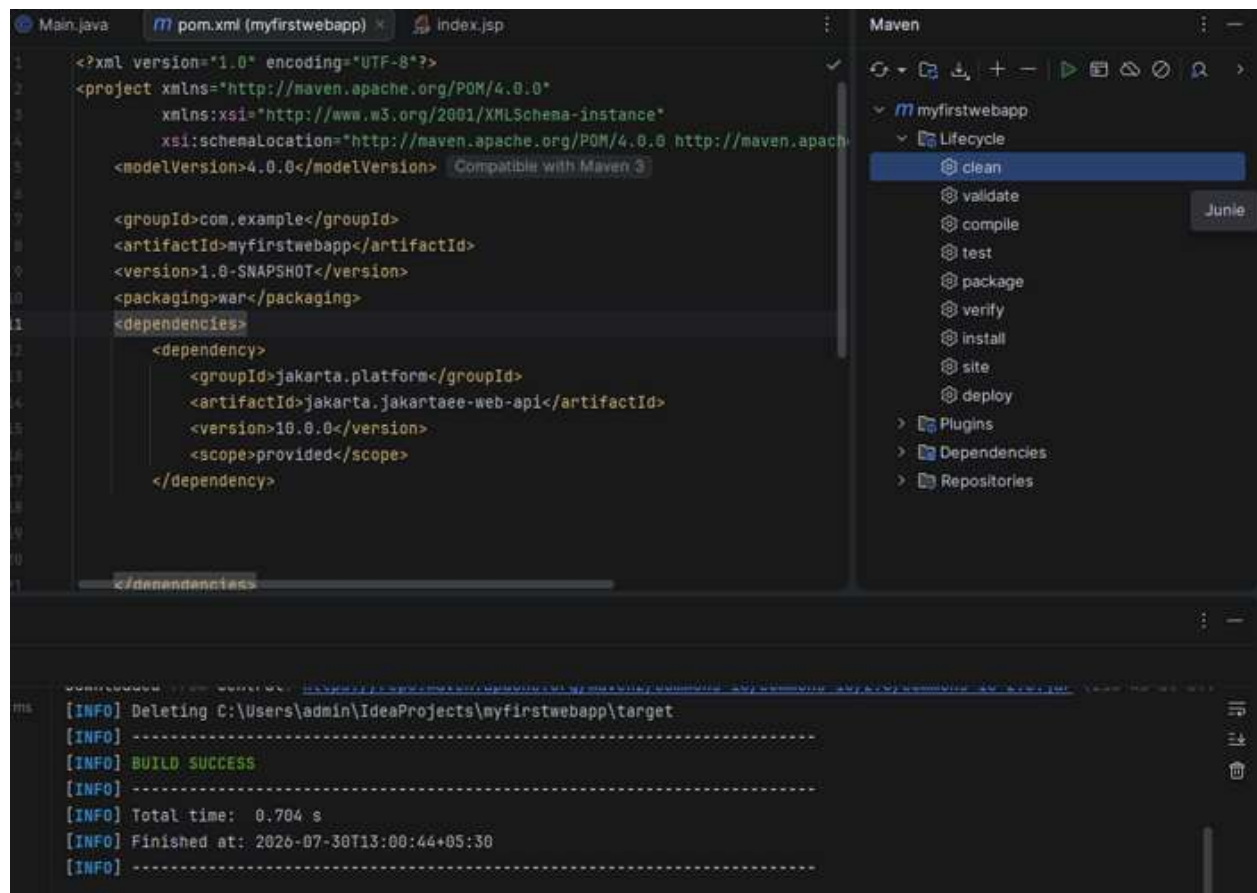
```
<!--
Created by IntelliJ IDEA.
User: admin
Date: 7/30/2026
Time: 12:13 PM
To change this template use File | Settings | File Templates.
-->
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<html>
<head>
    <title>My first web app</title>
</head>
<body>
<h1>Hello World!</h1>
</body>
```

Run Tomcat 10.1

Console Tomcat Access Log

```
20-261-2026 12:16:01.754 INFO [main] org.apache.catalina.core.StandardService.startInternal Starting service [catalina]
20-261-2026 12:16:01.755 INFO [main] org.apache.catalina.core.StandardEngine.startInternal Starting service engine: [Apache/2.0.62
20-261-2026 12:16:01.729 INFO [main] org.apache.catalina.startup.HostConfig.deployDescriptor Deploying deployment descriptor [C:\Users\adein\
20-261-2026 12:16:02.478 INFO [main] org.apache.catalina.startup.HostConfig.deployDescriptor Deployment of deployment descriptor [C:\Users\adein\
20-261-2026 12:16:02.584 INFO [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler ["http-nio-8080"]
20-261-2026 12:16:02.554 INFO [main] org.apache.catalina.startup.Catalina.start Server startup in 1405 milliseconds
http://localhost:8080/firstwebapp
```

pom.xml



Output:

← ↻ ⓘ localhost8081/webapp/

Hello World!

```
package com.example;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.io.IOException;

@WebServlet("/calculate")
public class Servlet extends HttpServlet {
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        int a=Integer.parseInt(request.getParameter("num1"));
        int b=Integer.parseInt(request.getParameter("num2"));
        String operation=request.getParameter("operation");

        try{
            String xml=SOAPClient.callService(a,b,operation);

            int start=xml.indexOf("<return>");
            start=xml.indexOf(">",start)+1;
            int end=xml.indexOf("</return>");

            String result=xml.substring(start,end).trim();
            request.setAttribute("result",result);
        }catch (Exception ex){
            request.setAttribute("result",ex.getMessage());
        }
        request.getRequestDispatcher("index.jsp").forward(request,response);
    }
}
```

SOAPClient.java

```

package com.example;

import java.io.*;
import java.net.HttpURLConnection;
import java.net.URL;
import java.nio.charset.StandardCharsets;

public class SOAPClient {
    public static String callService(int a, int b, String operation) throws
Exception {

        String soapRequest = "<?xml version=\"1.0\" encoding=\"UTF-8\"?>" +
            "<soapenv:Envelope " +
            "xmlns:soapenv=\"http://schemas.xmlsoap.org/soap/envelope/" +
            "xmlns:tns=\"http://webservice.example.com/" +
            "<soapenv:Header/>" +
            "<soapenv:Body>" +
            "<tns:add>" +
            "<arg0>"+a+"</arg0>" +
            "<arg1>"+b+"</arg1>" +
            "</tns:add>" +
            "</soapenv:Body>" +
            "</soapenv:Envelope>";

        URL url = new URL("http://localhost:8080/arithmetic");
        HttpURLConnection conn = (HttpURLConnection) url.openConnection();
        conn.setRequestMethod("POST");
        conn.setDoOutput(true);
        conn.setRequestProperty("SOAPAction", "");

        OutputStream os = conn.getOutputStream();
        os.write(soapRequest.getBytes());
        os.flush();
        BufferedReader br = new BufferedReader(new
InputStreamReader(conn.getInputStream()));
        String line;
        StringBuffer response = new StringBuffer();
        while ((line = br.readLine()) != null) {
            response.append(line);
        }
        br.close();
        return response.toString();
    }
}

```

Output:

localhost:8081/webapp/calculate

SOAP Calculator

number1

number2

Operation
Addition ▼

Result = 3

localhost:8081/webapp/calculate

SOAP Calculator

number1

number2

Operation
Addition ▼

Result = 2

localhost:8081/webapp/calculate

SOAP Calculator

number1

number2

Operation
multiplication ▼

Result = 8

SOAP Calculator

number1

8

number2

2

Operation

Addition ▼

Calculate

Result = 4.0